

ECON 1550: International Finance

National Income Accounting for  
Open Economies

# Announcements

- Problem Set 1 due today before midnight
- Solutions and Problem Set 2 posted right after
- Read pages 58–67 of textbook before ~~Wednesday~~ <sup>FRIDAY</sup> lecture
- Fill in survey for office hours and section (right now!)

# Office Hours and Section Availability

Please fill in your availability:



<https://www.when2meet.com/?34820140-APJfJ>

# Agenda

- Balance of payments account
- Exchange rate determination with asset approach

# Macro Review: GDP

**Gross Domestic Product (GDP)** measures the total value of:

- **Production:** All final goods and services produced within a country
- **Income:** All income earned from production within a country
- **Value added:** Sum of value added at each stage of production

All three approaches yield the same number.

# Closed Economy

$$Y = C + I + G$$

- All output is either consumed, invested, or purchased by government
- No trade with the rest of the world

# Open Economy with GDP

$$GDP = C + I + G + \underbrace{EX - IM}_{NX}$$

- $EX$  = Exports (domestic goods sold abroad)
- $IM$  = Imports (foreign goods purchased domestically)
- $NX$  = Net exports (trade balance)

# GDP and GNP

- **GDP:** Value of production *within a country's borders*
  - Regardless of who owns the factors of production
- **GNP:** Value of production by *a country's residents*
  - Regardless of where production takes place
- **Relationship:**

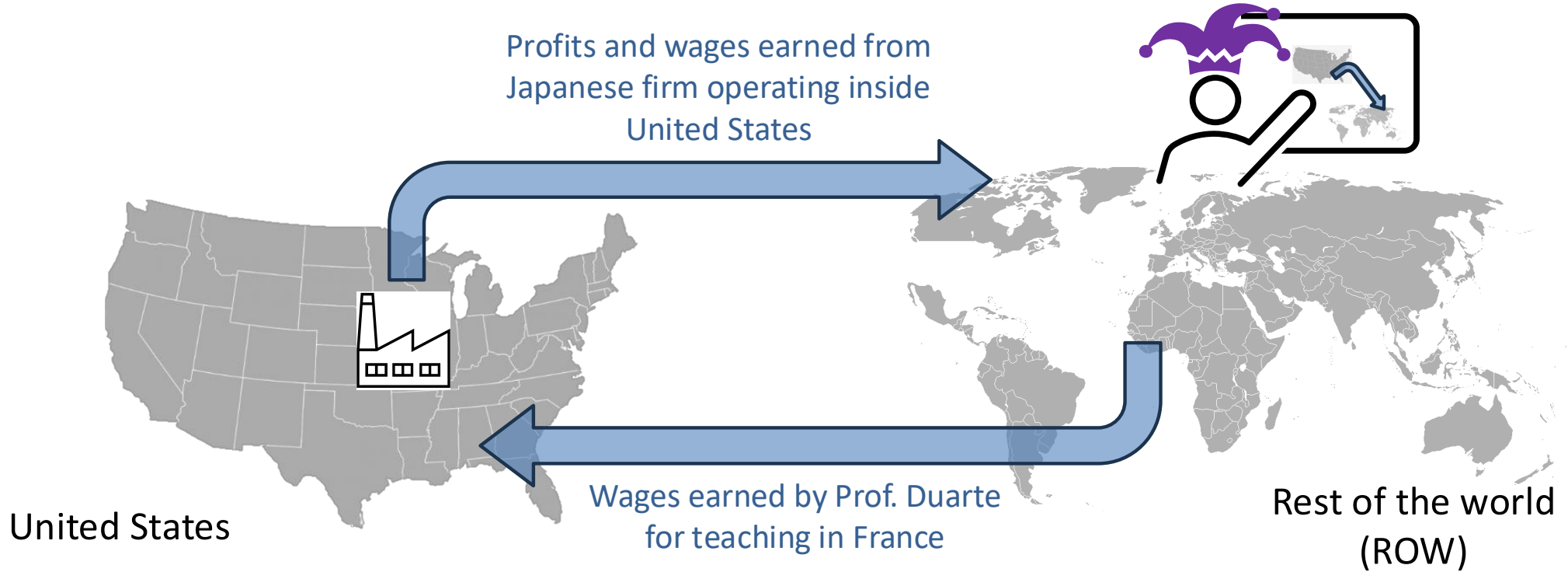
$$\text{GNP} = \text{GDP} + \text{Net Income from Abroad}$$

# Open Economy with GNP

$$GNP = C + I + G + CA$$

- $CA$  = Current account
- $CA = NX +$  Net income from abroad

# GDP and GNP



Net Income = wages from Prof. Duarte teaching in France — profits and wages from firm operating inside US

United States GNP = United States GDP + Net Income

# National Saving: Closed Economy

**National saving** = output not used for consumption or government spending

$$\begin{aligned} S &\equiv Y - C - G \\ &= (C + I + G) - C - G \\ &= I \end{aligned}$$

**In a closed economy:**  $S = I$

# National Saving: Open Economy

Starting from  $Y = C + I + G + NX$ :

$$S \equiv Y - C - G$$

$$= I + NX$$

**In an open economy:**  $S = I + NX$

# National Saving: Open Economy

Rearranging:  $S - I = NX$

- If  $S > I$ : Trade surplus, country is a net lender
- If  $S < I$ : Trade deficit, country is a net borrower

# Private Saving

**Private saving** = disposable income that is saved, not consumed

$$S^p \equiv Y - T - C$$

# Public (Government) Saving

**Government saving** = tax revenue minus government spending

$$S^g \equiv T - G$$

- If  $T > G$ : Budget surplus ( $S^g > 0$ )
- If  $T < G$ : Budget deficit ( $S^g < 0$ )

# Total Saving

**Total national saving:**

$$\begin{aligned} S &= S^p + S^g \\ &= (Y - T - C) + (T - G) \\ &= Y - C - G \end{aligned}$$

Using  $S = I + NX$  from before, we get:

$$S^p + S^g = I + NX$$

# Balance of Payments

|                          |      |
|--------------------------|------|
| <b>Current Account</b>   | -200 |
| <b>Capital Account</b>   | -1   |
| <b>Financial Account</b> | -201 |

→  $CA = NX + NI$

↪ NON-MARKET TRANSACTIONS

→ FINANCIAL  
FLOWS

$$\begin{array}{rclcl} \text{Current} & & \text{Capital} & & \text{Financial} \\ \text{Account} & + & \text{Account} & = & \text{Account} \\ (-200) & + & (-1) & = & -201 \end{array}$$

# Balance of Payments

|                                |          |                          |
|--------------------------------|----------|--------------------------|
| <b>Current Account</b>         | -200     | } -201                   |
| <b>Capital Account</b>         | -1       |                          |
| <b>Financial Account</b>       | -201     | } -201                   |
| <b>Statistical discrepancy</b> | <u>0</u> | ← "MEASUREMENT<br>Error" |

$$\begin{array}{rclcl}
 \text{Statistical} & = & \text{Financial} & - & \left( \begin{array}{cc} \text{Current} & \text{Capital} \\ \text{Account} & \text{Account} \end{array} \right) \\
 \text{Discrepancy} & & \text{Account} & & \\
 \hline
 \underline{0} & = & -201 & - & [ (-200) + (-1) ]
 \end{array}$$

## U.S. Balance of Payments Accounts for 2025-Q3 (\$ bn)

|                 |      |
|-----------------|------|
| Current Account | −226 |
|-----------------|------|

|                 |    |
|-----------------|----|
| Capital Account | −1 |
|-----------------|----|

|                   |      |
|-------------------|------|
| Financial Account | −410 |
|-------------------|------|

|                         |      |
|-------------------------|------|
| Statistical discrepancy | −183 |
|-------------------------|------|

$$\begin{array}{rclcl} \text{Statistical} & & \text{Financial} & & \\ \text{Discrepancy} & = & \text{Account} & - & \left( \begin{array}{c} \text{Current} \\ \text{Account} \end{array} + \begin{array}{c} \text{Capital} \\ \text{Account} \end{array} \right) \\ -183 & = & -410 & - & [ \begin{array}{c} (-226) \\ \end{array} + \begin{array}{c} (-1) \\ \end{array} ] \end{array}$$

Source: [BEA](#)

---

## Current Account

- |    |                                             |            |             |
|----|---------------------------------------------|------------|-------------|
| 1  | Exports total                               |            |             |
| 2  | Goods                                       |            |             |
| 3  | Services                                    |            |             |
| 4  | Income receipts (primary income)            | PART OF    |             |
| 5  | Imports total                               | NET INCOME | → LIKE      |
| 6  | Goods                                       |            | SERVICES    |
| 7  | Services                                    |            | PROVIDED BY |
| 8  | Income payments (primary income)            | PART OF    | AMERICAN    |
| 9  | Net unilateral transfers (secondary income) | NET INCOME | FACTORS TO  |
| 10 | <b>Balance on current account</b>           |            | ROW         |

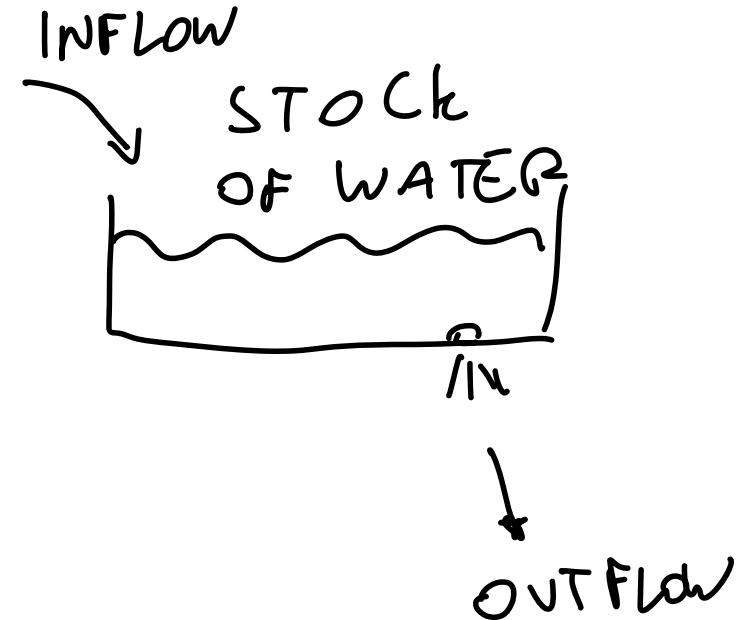
## Capital Account

- |    |                            |
|----|----------------------------|
| 11 | Balance on capital account |
|----|----------------------------|
-

---

## Financial Account

- 12 Net U.S. acquisition of foreign financial assets
  - 13 ✓ Official reserve assets → CENTRAL BANKS ✓
  - 14 Other assets → PRIVATE SECTOR
  - 15 Net U.S. incurrence of domestic liabilities
  - 16 Official reserve assets
  - 17 Other liabilities
  - 18 Financial derivatives, net
  - 19 **Net financial flows**  $(12) - (15) + (18)$
  - 20 **Statistical discrepancy**
- 



# Bottom line to remember

Whenever goods cross borders, assets move  
in the opposite direction.